

Chaos in the Lorenz System: The Butterfly Effect and the Lyapunov Exponent

1 The Lorenz System

The Lorenz system is a set of three coupled, nonlinear ordinary differential equations, originally derived by Edward Lorenz in 1963 as a drastically simplified model of atmospheric convection:

$$\dot{x} = \sigma(y - x) \tag{1}$$

$$\dot{y} = x(\rho - z) - y \tag{2}$$

$$\dot{z} = xy - \beta z \tag{3}$$

where $x, y, z \in \mathbb{R}$ are the state variables and $\sigma, \rho, \beta > 0$ are parameters. The classic chaotic regime uses

$$\sigma = 10, \quad \rho = 28, \quad \beta = \frac{8}{3}.$$

Despite being a smooth, deterministic, three-dimensional system, this choice of parameters produces trajectories that never repeat, remain bounded within a fractal set (the *attractor*), and are extraordinarily sensitive to initial conditions.

2 Numerical Integration: 4th-Order Runge–Kutta

Given a system $\dot{\mathbf{s}} = f(\mathbf{s})$ with state $\mathbf{s} = (x, y, z)$, the classic RK4 method advances the state by a time step h via

$$k_1 = f(\mathbf{s}_n) \tag{4}$$

$$k_2 = f\left(\mathbf{s}_n + \frac{h}{2}k_1\right) \tag{5}$$

$$k_3 = f\left(\mathbf{s}_n + \frac{h}{2}k_2\right) \tag{6}$$

$$k_4 = f(\mathbf{s}_n + h k_3) \tag{7}$$

$$\mathbf{s}_{n+1} = \mathbf{s}_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4) \tag{8}$$

This is fourth-order accurate: the local truncation error per step is $O(h^5)$, and the global error over a fixed time interval is $O(h^4)$. The implementation uses $h = 0.005$, which is small enough to keep the integration well within its stability region for the Lorenz parameters used here, while remaining fast enough to simulate tens of thousands of steps in a fraction of a second.

3 Sensitive Dependence on Initial Conditions

Let $\mathbf{s}_A(t)$ and $\mathbf{s}_B(t)$ be two trajectories of the same system with initial separation

$$\delta_0 = \|\mathbf{s}_B(0) - \mathbf{s}_A(0)\| = \varepsilon,$$

where ε is arbitrarily small (in this project, $\varepsilon = 10^{-8}$). For a chaotic system, the separation $\delta(t) = \|\mathbf{s}_B(t) - \mathbf{s}_A(t)\|$ grows, on average, *exponentially*:

$$\delta(t) \approx \delta_0 e^{\lambda t}$$

for as long as $\delta(t)$ remains small compared to the size of the attractor. The constant λ is the **largest Lyapunov exponent**. Rearranging,

$$\lambda \approx \frac{1}{t} \ln \left(\frac{\delta(t)}{\delta_0} \right).$$

- If $\lambda \leq 0$: nearby trajectories converge or stay together — the system is *not* chaotic.
- If $\lambda > 0$: nearby trajectories diverge exponentially — the system *is* chaotic, and $1/\lambda$ is a natural estimate of the “prediction horizon”: the timescale beyond which two initial conditions indistinguishable at measurement precision lead to unrelated outcomes.

For the classic Lorenz parameters, the widely cited reference value is

$$\lambda \approx 0.905 \text{ (per unit time).}$$

4 Benettin’s Algorithm

Naively integrating two trajectories ε apart and measuring $\delta(t)$ directly runs into a problem: $\delta(t)$ grows exponentially, so within a short time it saturates at the size of the attractor itself (here, tens of units) and the exponential-growth signal is lost in the nonlinear folding of the attractor. Benettin’s method (also called the *renormalization method*) fixes this:

1. Integrate a reference trajectory $\mathbf{s}_{\text{ref}}(t)$ and a perturbed trajectory $\mathbf{s}_{\text{pert}}(t)$, starting a distance ε apart, for a short interval of N steps.
2. Measure the new separation $d = \|\mathbf{s}_{\text{pert}} - \mathbf{s}_{\text{ref}}\|$.
3. Accumulate $\ln(d/\varepsilon)$ into a running sum.
4. **Rescale** $\mathbf{s}_{\text{pert}}$ back to distance exactly ε from $\mathbf{s}_{\text{ref}}$, *along the same direction of divergence*:

$$\mathbf{s}_{\text{pert}} \leftarrow \mathbf{s}_{\text{ref}} + \frac{\varepsilon}{d} (\mathbf{s}_{\text{pert}} - \mathbf{s}_{\text{ref}})$$

5. Repeat for many renormalization cycles $r = 1, \dots, M$, each covering a time interval Nh .

After M renormalizations covering total time $T = M \cdot Nh$, the largest Lyapunov exponent is estimated as

$$\lambda \approx \frac{1}{T} \sum_{r=1}^M \ln \left(\frac{d_r}{\varepsilon} \right).$$

This works because rescaling after every short interval keeps the perturbation small enough to stay in the locally linear regime around the reference trajectory (where the exponential-divergence approximation is valid), while the accumulated log-growth still correctly captures the average exponential rate over the whole attractor, including its folding and stretching.

Parameters used in this project

$$\varepsilon = 10^{-8}, \quad N = 10 \text{ steps/renorm}, \quad M = 4000 \text{ renorms}, \quad h = 0.005$$

giving a total integration time of $T = 4000 \times 10 \times 0.005 = 200$ time units, over which the running estimate of λ converges to

$$\lambda \approx 0.9032,$$

within 0.3% of the reference value 0.905.

5 Summary

Quantity	Value / Method
System	Lorenz equations, $\sigma = 10$, $\rho = 28$, $\beta = 8/3$
Integrator	RK4, step size $h = 0.005$
Chaos test	Two trajectories, $\varepsilon = 10^{-8}$ apart
Divergence	Exponential: $\delta(t) \approx \varepsilon e^{\lambda t}$
Lyapunov exponent	$\lambda \approx 0.903$ (computed) vs. 0.905 (reference)
Prediction horizon	$1/\lambda \approx 1.1$ time units